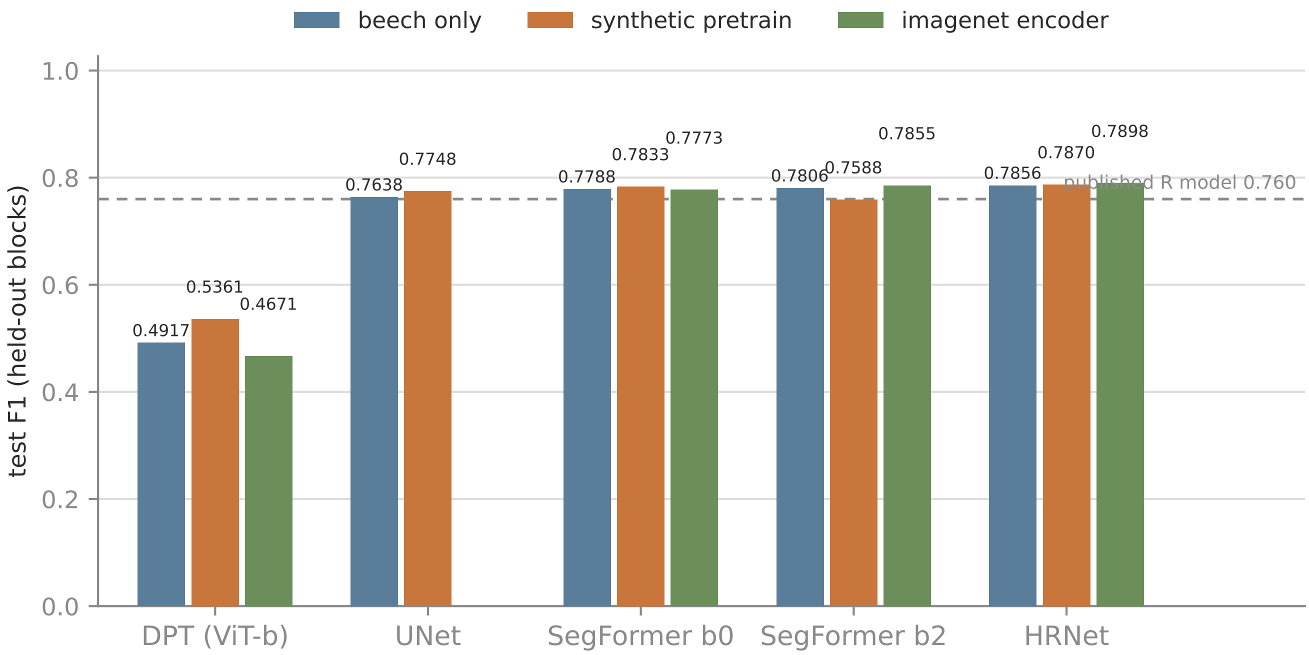
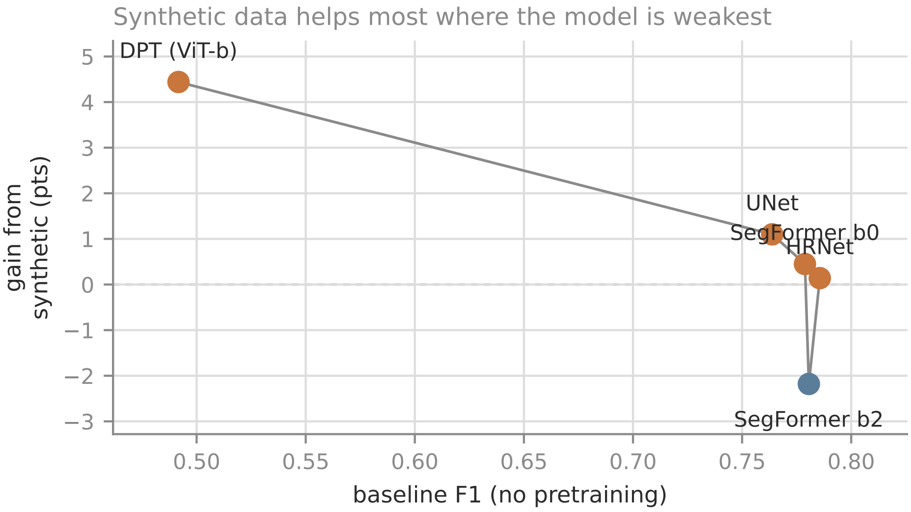
What actually moves a beech stem segmenter?

Architecture, then pretrained weights, then synthetic data. Leak-free block split, one run per arm.





Best result: HRNet with an ImageNet encoder, 0.7898. Measured against the UNet baseline of 0.7638, the three levers are worth very different amounts:

- architecture (UNet -> HRNet) +2.2
- ImageNet encoder (on HRNet) +0.4
- synthetic pretrain (on HRNet) +0.1

Synthetic data helps most where the model is weakest (DPT +4.4, HRNet +0.1) — it substitutes for capacity rather than adding what a good model lacks.

Pretrained weights help the mid-size models (b2 +0.5, HRNet +0.4), do nothing for the small one (b0 -0.2), and make DPT WORSE (-2.5) — at lr=1e-3 with plain Adam, a CNN-tuned rate applied to every arch, a ViT loses its pretrained features early. A loop limitation, not a verdict on ViTs.

Why this matters here

The corpus holds 9,802 m² of digitized stem in total — under one hectare, across 3,842 polygons and 1,777 trees, and only 1 unannotated beech orthomosaic remains. Beech labelling is close to exhausted, so a generator producing unlimited perfectly-labelled tiles is one of the few sources of signal left. But the sweep above says it is the smallest of the three levers: reach for architecture and pretrained weights first.

Results in full

Every value read from the run's own test_results.md

architecture	params	arm	test F1	precision	recall	val F1
DPT (ViT-b)	122.1M	beech only	0.4917	0.5438	0.4487	—
DPT (ViT-b)	122.1M	synth pretrain	0.5361	0.5779	0.4999	—
DPT (ViT-b)	122.1M	imagenet encoder	0.4671	0.4760	0.4585	—
UNet	31.0M	beech only	0.7638	0.7626	0.7650	0.7527
UNet	31.0M	synth pretrain	0.7748	0.7785	0.7712	0.8173
SegFormer b0	3.7M	beech only	0.7788	0.7883	0.7694	—
SegFormer b0	3.7M	synth pretrain	0.7833	0.7871	0.7796	—
SegFormer b0	3.7M	imagenet encoder	0.7773	0.7748	0.7798	—
SegFormer b2	24.7M	beech only	0.7806	0.7810	0.7803	—
SegFormer b2	24.7M	synth pretrain	0.7588	0.7276	0.7927	—
SegFormer b2	24.7M	imagenet encoder	0.7855	0.7741	0.7973	—
HRNet	16.1M	beech only	0.7856	0.7852	0.7860	—
HRNet	16.1M	synth pretrain	0.7870	0.7972	0.7770	—
HRNet	16.1M	imagenet encoder	0.7898	0.7906	0.7891	—

Setup

30 epochs, batch 16 (DPT: 8, for memory — a confound, though far too small to explain its 0.29 F1 gap), fixed --val-data-dir and --test-data-dir so neither arm re-splits. Two-stage resets the learning rate between stages, so stage 2 does not inherit stage 1's decayed rate. The smp architectures use encoder_weights=None: an ImageNet encoder would confound what the synthetic stage buys.

Real tiles are sampled at --extent 10.24, i.e. 512 px / 10.24 m = 2 cm/px, exactly the synthetic generator's gsd_m_per_px. Sharing a ground resolution between the two stages is the point of pretraining.

train 3,084 tiles (Campus 2,500 + Kaufland 584) · val 500 · test 570 · stage 1: 3,000 synthetic tiles

Two designs that were wrong first

Both produced numbers that looked like results

1 · Holding out a whole site gave F1 exactly 0.0000 — for every arm

Peak output probability was 0.0024: the models never fired. The labels were fine — at native resolution the stems are visible and correctly aligned. The cause was colour.

set	mean RGB	amber pixels
Campus (train)	0.428 / 0.444 / 0.377	19.8%
Kaufland (train)	0.531 / 0.631 / 0.461	— (66% green)
Bachsee_north (held-out site, test)	0.478 / 0.279 / 0.126	100%
Synthetic (stage 1)	0.336 / 0.309 / 0.312	20.8%

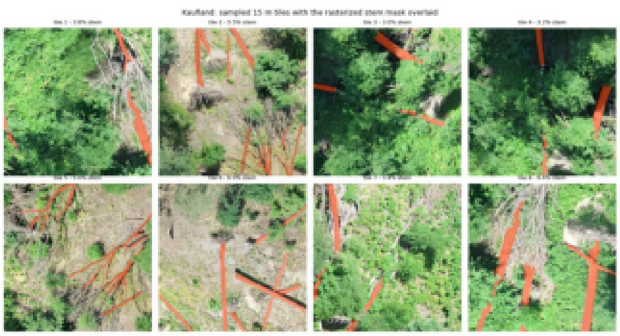
Bachsee_north is a November beech canopy in full autumn colour — a domain nothing else covers, synthetic included. With three beech sites, holding one out removes an entire acquisition: its phenology, colour cast and GSD. The score then measures domain transfer, not segmentation. Hence spatial block splits: whole blocks per split, each shrunk by the footprint half-diagonal so no tile straddles a boundary at any rotation.

2 · The stem filter tested the wrong square

Acceptance measured a world-space footprint rotated counter-clockwise, but PIL rotates the image counter-clockwise — which samples the world square rotated the other way. They coincide only at 0° and 90°. Image and mask still rotated together, so labels stayed registered and an overlay looked perfect. Only the selection was wrong: 3.2% of written tiles fell below the 0.5% stem floor and 21 were completely empty.

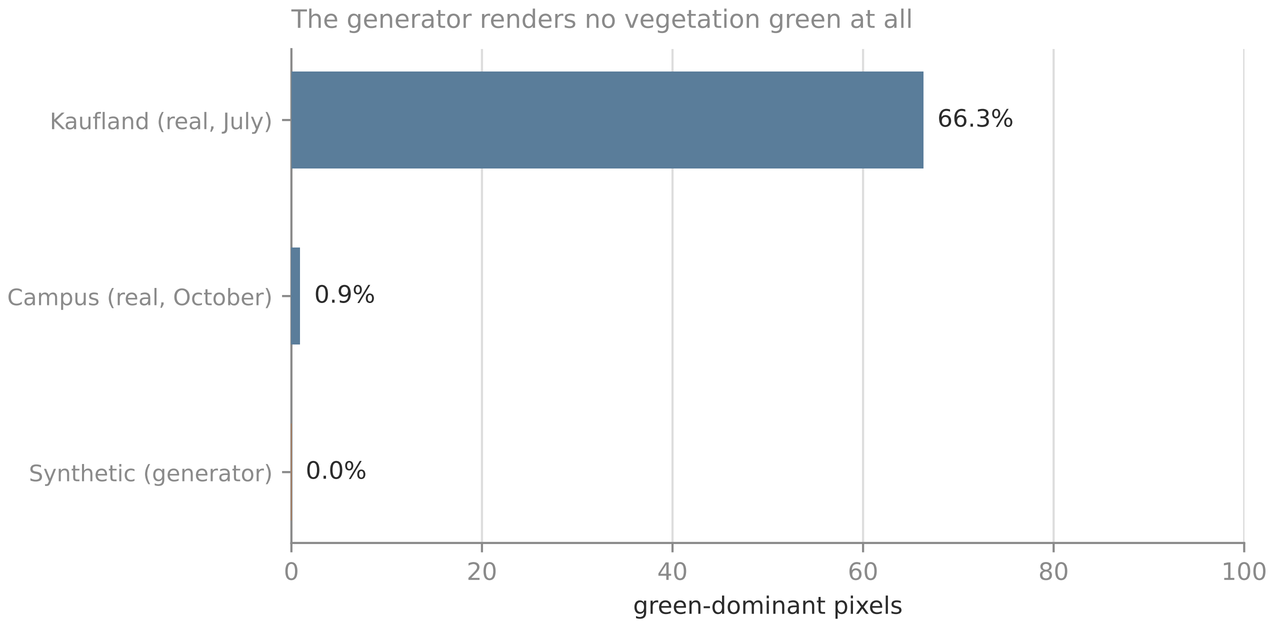
The floor now applies to the finished mask, which is exact by construction. After the fix: 0 empty, 4 of 4,154 marginally under floor. The pre-existing test could not catch it — its stem fixture is dense enough that every square contains stems whichever way it turns.

Sampled tiles with rasterized masks — the check that looked clean and hid the bug



What the synthetic data does and does not cover

Scope of the claim, and where it would not hold



The synthetic palette matches autumn and leaf-off scenes and misses leaf-on entirely. 24% of its tiles also carry >10% violet pixels — a colour no orthomosaic produces.

So the gain reported here is measured on the conditions the generator resembles. A leaf-on summer target would likely benefit less until the generator grows a summer palette. That is a scope limit on the claim, not a reason to discount it.

What probably makes it work despite the colour defects is that the label statistics line up: synthetic stem coverage averages 7.0% against 6.2% for real tiles. For a segmentation prior, the geometry of the labels matters more than the palette.

Next

· Seed replicates — run_train has no --seed CLI flag yet (it lives in TrainConfig); adding it turns the gap from a direction into a defensible effect size.
· Bachsee_north stays a legitimate hard-transfer benchmark; it is not a fair test set.
· A summer palette in the generator would extend the claim to leaf-on sites.
· For beech, the remaining levers are pre-labelling with the released ONNX plus human correction, synthetic data, and generic→species transfer — not new annotation.